

Exercise 1 (5 points)

Part 1 (3 points)

We consider the following expression:

$$u(x) = \frac{1}{\sqrt{x^2 + 1} - x}$$

1. Compute u for $x = 555$ using three (3) significant digits with chopping.

0.25 $x^2 = (555)^2 = 308000$

0.25 $x^2 + 1 = 308000 + 1 = 308000$

0.25 $\sqrt{x^2 + 1} = 554$

0.25 $\sqrt{x^2 + 1} - x = 554 - 555 = -1$ $u(555) = \frac{1}{-1} = -1$

2. Compare with the exact value (compute the relative error).

0.5 $exacte = 1110.001$

0.5 $e_{rel} = \left| \frac{u_{num} - u_{ex}}{u_{num}} \right| = 100\%$

3. What do you observe? Explain your results and propose a solution to the problem.

1 we observe that the approximation obtained is completely false, and even worse because we have a sing error. Explanation: the subtraction of two number Almost equal $\sqrt{x^2 + 1}$ and x caused this problem (after chopping in x^2 , $x^2 + 1$, and $\sqrt{x^2 + 1}$)

Part 2 (2 points)

A computer uses base 10 with $\epsilon_{mach} = 10^{-3}$ and an exponent range $-8 < e < 6$. Give the result of each of the following operations if they are performed on this computer:

0.25 1. $457 + 1 = 457 \cdot (1 + \frac{1}{457}) = 458$

0.25 2. $14.697 \times 10^3 = 1.4697 \times 10^4$

0.5 3. $\ln(10^6 + 2) = \ln(10 + 2) = 10$ (overflow)

0.5 4. $\frac{2}{10^{-10}(10^{10} + 1)} = \frac{2}{0(\infty)} = \text{NaN (Not a Number)}$

0.5 5. $\frac{10^{11}}{10^8} = \frac{\text{inf}}{\text{inf}} = \text{Not a Number (Indeterminate form)}$

Full Name :

Group:

Exercise 2 (8 points)

The equation

$$f(x) = x + e^{-x} - 4 = 0$$

admits a unique negative root $x^* \approx -1.7$. Consider the two functions

$$g_1(x) = -\ln(4-x), \quad g_2(x) = 4 - e^{-x}.$$

1. Verify that each equation $x = g_i(x)$ is equivalent to $f(x) = 0$.

0.5 $x = g_1(x) \Rightarrow x = -\ln(4-x) \Rightarrow -x = \ln(4-x) \Rightarrow e^{-x} = 4-x \Rightarrow f(x) = 0$

0.5 $x = g_2(x) \Rightarrow x = 4 - e^{-x} \Rightarrow x + e^{-x} - 4 = 0 \Rightarrow f(x) = 0$

2. Among the two functions g_1 and g_2 , determine which one can be used in the fixed-point iteration $x_{n+1} = g_i(x_n)$ to compute the root x^* of $f(x) = 0$.

1 $g'_1(x) = \frac{1}{4-x} \Rightarrow |g'_1(x^*)| \approx \left| \frac{1}{4+1.7} \right| \approx 0.17 < 1 \Rightarrow \text{Contractive} \Rightarrow \text{Converges}$

1 $g'_2(x) = e^{-x} \Rightarrow |g'_2(x^*)| \approx e^{1.7} \approx 5.47 > 1 \Rightarrow \text{Not Contractive} \Rightarrow \text{Diverge}$

0.5 so we use: $x_{n+1} = g_1(x_n)$

3. Using the iteration chosen in the previous question and 4 significant digits with rounding, compute x_2 starting from $x_0 = -1.8$. Complete the following table.

Iteration (n)	x_n	$g(x_n)$	$ x_n - x_{n-1} $
0	-1.8	$-\ln(4+1.8) = -1.758$	0.042 0.042
1	-1.758	$-\ln(4-x_1) = -1.751$	0.007

How many significant digits are correct? (Give the convergence error.)

There are 3 significant digits correct (between x_1 and x_2) and error of convergence est =

$e_1 = 0.007$

4. Determine the number of iterations required so that the relative error achieves 0.0001

$|x_n - x^*| \leq \frac{k^n}{1-k} |x_1 - x_0| \Rightarrow \frac{(0.175)^n}{1-0.175} \times 0.042 \leq 0.0001$

$(0.175)^n \times 0.0509 \leq 0.0001 \Rightarrow 0.175^n \leq 0.00196$

$n \cdot \ln(0.175) \leq \ln(0.00196)$

$n \geq \frac{-6.23}{-1.74} \approx 3.58$

$n = 4$

Exercise 3 (7 points)

The exact solution of the first linear system $Ax = b$ below is

$$x^* = \begin{bmatrix} 10 \\ 20 \end{bmatrix}.$$

1. Using three (3) significant digits with rounding, complete the two Gaussian eliminations shown below. Give the numerical solution x_{num} obtained in each case.

A **b**

$$\left[\begin{array}{cc|c} 0.01 & 1.6 & 32.1 \\ 1 & 0.6 & 22 \end{array} \right] \xrightarrow{R_2 - 100 R_1} \left[\begin{array}{cc|c} 0.01 & 1.6 & 32.1 \\ 0 & -1.59 & -31.90 \end{array} \right]$$

Solve $Ax = b$... $0.6 - 100 \times (1.6) = -159$ $22 - 100 \times (32.1) = -3198 \approx -3190$ (3 sig digits)

eq 2: $-1.59 x_2 = -31.90 \Rightarrow x_2 \approx 20.1$

eq 1: $0.01 x_1 + 1.6(20.1) = 32.1 \Rightarrow 0.01 x_1 = -0.1 \Rightarrow x_1 \approx -10$

$x^* = \begin{bmatrix} -10 \\ 20.1 \end{bmatrix}$ (1)

$$\left[\begin{array}{cc|c} 1 & 0.6 & 22 \\ 0.01 & 1.6 & 32.1 \end{array} \right] \xrightarrow{R_2 - 0.01 R_1} \left[\begin{array}{cc|c} 1 & 0.6 & 22 \\ 0 & 1.59 & 31.9 \end{array} \right]$$

Solve $Ax = b$... $1.6 - 0.01(0.6) = 1.594 \approx 1.59$ $32.1 - 0.01(22) = 32.1 - 0.22 = 31.88 \approx 31.9$

eq 2: $1.59 x_2 = 31.9 \Rightarrow x_2 \approx 20.1$

eq 1: $x_1 + 0.6(20.1) = 22 \Rightarrow x_1 = 22 - 12.1 = 9.9$

$x^* = \begin{bmatrix} 9.9 \\ 20.1 \end{bmatrix}$ (1)

2. What do you observe? Explain your results.

x^* is completely false (100% error), x^* is very accurate (2) (rel $\approx 0.5\%$). This is explained by the small pivot 0.01, which is replaced by 1 after partial pivoting.

3. Give the factorization $A = LU$.

(1) $A = \begin{bmatrix} 1 & 0 \\ 100 & 1 \end{bmatrix} \begin{bmatrix} 0.01 & 1.6 \\ 1.59 & -1.59 \end{bmatrix}$ Even if the solution obtained by Gaussian Elimination without pivoting is wrong, the LU decomposition itself is still valid.

===== Best of Luck =====